ClipSAT : ACT International Subject Test: Mathematics Level 2 [ANSWER KEY]

Unit 2.01: Matrices, Determinants & Vectors

Q1. Given the vectors mathbf{u} = langle 3, -4 rangle and mathbf{v} = langle -2, 5 rangle , what is the vector 2mathbf{u} + mathbf{v} ?

[x] A: langle 4, -3 rangle

[ ] B: langle 1, 1 rangle

[ ] C: langle 4, 1 rangle

[ ] D: langle 8, -13 rangle

Solution: To find 2mathbf{u} + mathbf{v} , first perform the scalar multiplication on mathbf{u} : 2mathbf{u} = 2langle 3, -4 rangle = langle 23, 2-4 rangle = langle 6, -8 rangle Next, add the corresponding components of 2mathbf{u} and mathbf{v} : 2mathbf{u} + mathbf{v} = langle 6 + -2, -8 + 5 rangle = langle 4, -3 rangle - langle 1, 1 rangle is incorrect because it represents mathbf{u} + mathbf{v} , omitting the scalar factor of 2. - langle 4, 1 rangle is incorrect because the scalar 2 was only distributed to the horizontal component. - langle 8, -13 rangle is incorrect because it calculates 2mathbf{u} - mathbf{v} instead of 2mathbf{u} + mathbf{v} .

Q2. Matrices A and B are defined as A = begin{bmatrix} 2 & k -1 & 3 end{bmatrix} and B = begin{bmatrix} 1 & 4 2 & k end{bmatrix} , where k is a real number. If the determinant of the matrix sum A + B is equal to 17 , what is the value of k ? Note: The determinant of a 2 times 2 matrix begin{bmatrix} a & b c & d end{bmatrix} is defined as ad - bc .

[ ] A: 1

[ ] B: 2

[x] C: 6

[ ] D: frac{19}{2}

Solution: Step 1: Compute the matrix sum A + B by adding corresponding entries: A + B = begin{bmatrix} 2 + 1 & k + 4 -1 + 2 & 3 + k end{bmatrix} = begin{bmatrix} 3 & k + 4 1 & k + 3 end{bmatrix} Step 2: Calculate the determinant of A + B using the definition detbegin{bmatrix} a & b c & d end{bmatrix} = ad - bc : detA + B = 3k + 3 - 1k + 4 Distribute carefully across both terms: detA + B = 3k + 9 - k - 4 = 2k + 5 Step 3: Set the determinant equal to 17 and solve for k : 2k + 5 = 17 implies 2k = 12 implies k = 6 Common Misconceptions: - Choice B 2 results from a sign distribution error when subtracting: 3k + 9 - k + 4 = 2k + 13 = 17 implies k = 2 . - Choice A 1 results from incorrectly using ad + bc instead of ad - bc : 3k + 9 + k + 4 = 4k + 13 = 17 implies k = 1 . - Choice D frac{19}{2} comes from the false assumption that detA + B = detA + detB , which gives 6 + k + k - 8 = 2k - 2 = 17 implies k = frac{19}{2} .

Q3. For all real values of theta, the matrix product begin{bmatrix} costheta & sintheta end{bmatrix} begin{bmatrix} 6 & 2 2 & 3 end{bmatrix} begin{bmatrix} costheta sintheta end{bmatrix} evaluates to a 1 times 1 matrix [k]. What is the maximum possible value of k?

[ ] A: 6

[x] B: 7

[ ] C: 9

[ ] D: 14

Solution: Step 1: Perform the matrix multiplication. First, multiply the 1 times 2 row vector by the 2 times 2 matrix: begin{bmatrix} costheta & sintheta end{bmatrix} begin{bmatrix} 6 & 2 2 & 3 end{bmatrix} = begin{bmatrix} 6costheta + 2sintheta & 2costheta + 3sintheta end{bmatrix} Next, multiply this result by the 2 times 1 column vector: k = 6costheta + 2sinthetacostheta + 2costheta + 3sinthetasintheta k = 6cos^2theta + 4sinthetacostheta + 3sin^2theta Step 2: Express the quadratic trigonometric terms using double-angle identities. Recall the standard identities: cos^2theta = frac{1 + cos2theta}{2}, quad sin^2theta = frac{1 - cos2theta}{2}, quad 2sinthetacostheta = sin2theta Substitute these into the expression for k: k = 6leftfrac{1 + cos2theta}{2}right + 3leftfrac{1 - cos2theta}{2}right + 2sin2theta k = 31 + cos2theta + frac{3}{2}1 - cos2theta + 2sin2theta k = frac{9}{2} + frac{3}{2}cos2theta + 2sin2theta Step 3: Maximize the resulting trigonometric expression. For real constants A and B, the maximum value of Acosphi + Bsinphi is sqrt{A^2 + B^2}. Here, A = frac{3}{2} and B = 2: sqrt{leftfrac{3}{2}right^2 + 2^2} = sqrt{frac{9}{4} + frac{16}{4}} = sqrt{frac{25}{4}} = frac{5}{2} Thus, the maximum value of k is: k_{text{max}} = frac{9}{2} + frac{5}{2} = 7 Misconception Analysis: - Choice A 6 results from ignoring the cross-product term 4sinthetacostheta and merely selecting the maximum diagonal entry. - Choice C 9 results from taking the trace sum

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Q4. Given the vectors mathbf{u} = langle 3, -2 rangle and mathbf{v} = langle 1, 5 rangle , what is the vector 2mathbf{u} + mathbf{v} ?

[x] A: langle 7, 1 rangle

[ ] B: langle 4, 3 rangle

[ ] C: langle 8, 6 rangle

[ ] D: langle 5, 8 rangle

Solution: To find 2mathbf{u} + mathbf{v} , first perform the scalar multiplication on mathbf{u} : 2mathbf{u} = 2langle 3, -2 rangle = langle 23, 2-2 rangle = langle 6, -4 rangle Next, add vector mathbf{v} by adding corresponding components: 2mathbf{u} + mathbf{v} = langle 6, -4 rangle + langle 1, 5 rangle = langle 6 + 1, -4 + 5 rangle = langle 7, 1 rangle Common Errors: - langle 4, 3 rangle results from computing mathbf{u} + mathbf{v} without multiplying mathbf{u} by 2. - langle 8, 6 rangle results from multiplying both vectors by 2, computing 2mathbf{u} + mathbf{v} . - langle 5, 8 rangle results from applying the scalar 2 to mathbf{v} instead of mathbf{u} , computing mathbf{u} + 2mathbf{v} .

Q5. For a 2 times 2 matrix begin{bmatrix} a & b c & d end{bmatrix}, the determinant is defined as ad - bc. Given the matrices A = begin{bmatrix} x & 2 -1 & 3 end{bmatrix} and B = begin{bmatrix} 1 & -4 3 & x end{bmatrix}, the determinant of the matrix A + B is equal to 19. If x > 0, what is the value of x?

[x] A: 2

[ ] B: frac{5}{4}

[ ] C: 4

[ ] D: 6

Solution: First, find the matrix sum A + B by adding corresponding entries: A + B = begin{bmatrix} x + 1 & 2 + -4 -1 + 3 & 3 + x end{bmatrix} = begin{bmatrix} x + 1 & -2 2 & x + 3 end{bmatrix} Next, calculate the determinant of A + B: detA + B = x + 1x + 3 - -22 detA + B = x^2 + 4x + 3 - -4 = x^2 + 4x + 7 Set the determinant equal to 19 and solve for x: x^2 + 4x + 7 = 19 x^2 + 4x - 12 = 0 Factoring the quadratic gives: x + 6x - 2 = 0 This yields x = -6 or x = 2. Since we are given that x > 0, the value is x = 2. Common Pitfalls: - Assuming detA + B = detA + detB: detA = 3x + 2 and detB = x + 12, so setting 4x + 14 = 19 results in the incorrect answer x = frac{5}{4}. - Factoring sign error: solving x^2 + 4x - 12 = 0 as x - 6x + 2 = 0 yields x = 6.

Q6. Let mathbf{u} and mathbf{v} be vectors in the coordinate plane such that |mathbf{u}| = 5, |mathbf{v}| = 10, and the angle theta between them is obtuse 90^circ < theta < 180^circ. If the minimum value of |tmathbf{u} + mathbf{v}| over all real numbers t is equal to 6, what is the value of the dot product mathbf{u} cdot mathbf{v}?

[ ] A: -50

[x] B: -40

[ ] C: -30

[ ] D: 40

Solution: We can solve this problem algebraically or geometrically. **Method 1: Algebraic Quadratic Optimization** Consider the square of the magnitude |tmathbf{u} + mathbf{v}|: |tmathbf{u} + mathbf{v}|^2 = tmathbf{u} + mathbf{v} cdot tmathbf{u} + mathbf{v} = t^2|mathbf{u}|^2 + 2tmathbf{u} cdot mathbf{v} + |mathbf{v}|^2 Let D = mathbf{u} cdot mathbf{v}. Substituting |mathbf{u}| = 5 and |mathbf{v}| = 10: ft = 25t^2 + 2Dt + 100 This is a quadratic function in t with leading coefficient a = 25 > 0. Its minimum occurs at the vertex: t_{min} = -frac{2D}{225} = -frac{D}{25} Evaluating ft at the vertex gives the minimum squared magnitude: ft_{min} = 25left-frac{D}{25}right^2 + 2Dleft-frac{D}{25}right + 100 = 100 - frac{D^2}{25} We are given that the minimum value of |tmathbf{u} + mathbf{v}| is 6, so the minimum squared magnitude is 6^2 = 36: 100 - frac{D^2}{25} = 36 implies frac{D^2}{25} = 64 implies D^2 = 1600 Taking the square root gives D = pm 40. Because the angle theta between mathbf{u} and mathbf{v} is obtuse 90^circ < theta < 180^circ, cos theta < 0, which implies mathbf{u} cdot mathbf{v} = |mathbf{u}||mathbf{v}|cos theta < 0. Therefore: mathbf{u} cdot mathbf{v} = -40 **Method 2: Geometric Orthogonal Distance** The set of points with position vectors tmathbf{u} + mathbf{v} forms a straight line parallel to mathbf{u} passing through the endpoint of mathbf{v}. The minimum magnitude |tmathbf{u} + mathbf{v}| is the perpendicular distance from the origin to this line. In the right triangle formed by the origin, the endpoint of

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Q7. What is the value of the determinant of the matrix begin{bmatrix} 7 & -3 2 & 4 end{bmatrix} ?

[x] A: 34

[ ] B: 22

[ ] C: -29

[ ] D: -34

Solution: The determinant of a 2 times 2 matrix begin{bmatrix} a & b c & d end{bmatrix} is given by the formula: det = ad - bc Substitute the entries a = 7 , b = -3 , c = 2 , and d = 4 : det = 74 - -32 = 28 - -6 = 28 + 6 = 34 Common misconceptions: - Calculating ad + bc gives 28 + -6 = 22 . - Reversing the subtraction order as bc - ad gives -6 - 28 = -34 . - Multiplying along the rows instead of diagonals gives 7-3 - 24 = -21 - 8 = -29 .

Q8. In the standard coordinate plane, vector mathbf{u} = langle 2, -1 rangle and vector mathbf{v} = langle k, 5 rangle , where k is a real constant. If the magnitude of the vector mathbf{u} + mathbf{v} is equal to 5, which of the following is a possible value of k ?

[x] A: -5

[ ] B: -1

[ ] C: 3

[ ] D: 7

Solution: First, find the sum of vectors mathbf{u} and mathbf{v} by adding their corresponding components: mathbf{u} + mathbf{v} = langle 2 + k, -1 + 5 rangle = langle 2 + k, 4 rangle Next, use the magnitude formula |langle x, y rangle| = sqrt{x^2 + y^2} : sqrt{2 + k^2 + 4^2} = 5 Square both sides of the equation to eliminate the radical: 2 + k^2 + 16 = 25 2 + k^2 = 9 Take the square root of both sides: 2 + k = 3 quad text{or} quad 2 + k = -3 k = 1 quad text{or} quad k = -5 Among the choices, only -5 is listed. Common traps: - Solving 2 + k + 4 = 5 by mistakenly adding components linearly instead of using the Pythagorean theorem yields k = -1 . - Forgetting the negative square root 2 + k = -3 causes students to miss k = -5 .